

Figure 7-2 Effect of changing interest rates on option values.

	If domestic rates rise	If domestic rates fall	If foreign rates rise	If foreign rates fall
stock option calls will	rise	fall	not applicable	not applicable
stock option puts will	fall	rise	not applicable	not applicable
futures option calls (stock-type settlement)	fall	fall	not applicable	not applicable
futures option puts (stock-type settlement)	fall	fall	not applicable	not applicable
futures option calls (futures-type settlement)	no	no	not applicable	not applicable
futures option puts (futures-type settlement)	effect	effect	not applicable	not applicable
foreign currency option calls	rise	fall	fall	rise
foreign currency option puts	fall	rise	rise	fall

Figure 7-3 Effect of changing dividends on stock option values.

	If the dividend is raised	If the dividend is cut
stock option calls will	fall	rise
stock option puts will	rise	fall

the *partial derivatives*, will not answer all our questions concerning changing market conditions, but they are an important starting point in analyzing the risks associated with both simple and complex option positions.

The Delta

The *delta* (Δ) is a measure of an option's risk with respect to the direction of movement in the underlying contract. A positive delta indicates a desire for upward movement; a negative delta indicates a desire for downward movement. The delta has several different interpretations, any of which may be useful to a trader depending on the types of strategies being executed.

Rate of Change

At expiration, an option is worth exactly its intrinsic value. Prior to expiration, however, the theoretical value of an option is a curve that will approach intrinsic value as the option goes very deeply into the money or very far out of the